Study 1b Analysis

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Primary model. For each foundation family, a logistic regression predicts whether a post invokes it (any individualizing; any binding) from framing category and sentiment. Both families are estimated jointly in one model. **Shift** compares opponent-directed with own-side posts within a side. **Convergence asymmetry** is pro-choice convergence toward the opponent's characteristic family minus pro-life convergence; positive means pro-choice users converge more. SEs are clustered by the `user` column. Those IDs are rounded in the file, which merges some accounts into one cluster, so the SEs are conservative.

The original multinomial model is reproduced and reported with correctly labeled SEs, alongside the recoding that keeps posts invoking both families separate.

```
suppressPackageStartupMessages({ library(nnet); library(dplyr); library(tidyr);
  library(ggplot2); library(sandwich) })
p <- params; OUT <- p$out_dir; dir.create(OUT, showWarnings = FALSE)
TXT <- file.path(OUT, "study1_results.txt"); cat("STUDY 1 FINAL RESULTS\n", file = TXT)
say <- function(...) { t <- paste0(...); cat(t, "\n", file = TXT, append = TRUE, sep = "")
; cat(t, "\n", sep = "") }
block <- function(title, df) { say("\n### ", title); o <- options(width = 220);
on.exit(options(o))
  t <- capture.output(print(as.data.frame(df), row.names = FALSE));
  cat(t, sep = "\n", file = TXT, append = TRUE); cat(t, sep = "\n") }
f_d <- function(x, d = 2) formatC(x, format = "f", digits = d)
f_pct <- function(x, d = 1) formatC(100 * x, format = "f", digits = d)
f_num <- function(n) format(n, big.mark = ",", trim = TRUE)
f_p <- function(p) ifelse(p < .001, "< .001", paste("=", sub("^0", "", sprintf("%.3f", p))))
f_or <- function(b, se) sprintf("%s [%s, %s]", f_d(exp(b)),
                                f_d(exp(b - 1.96 * se)), f_d(exp(b + 1.96 * se)))
t_num <- function(n) gsub(",", "{,}", f_num(n))
t_p <- function(p) ifelse(p < .001, "$<$.001", sub("^0", "", sprintf("%.3f", p)))
write_tex <- function(df, file, caption, label, note, align = NULL, size = "\\small") {
  df <- as.data.frame(df); df[] <- lapply(df, function(v)
    gsub("(^|\\[|\\(| )-(?=[0-9.])", "\\1$-$", as.character(v), perl = TRUE))
  if (is.null(align)) align <- paste0("l", strrep("r", ncol(df) - 1))
  writeLines(c("\\begin{table}[!htbp]", "\\centering", size, "\\begin{threeparttable}",
               paste0("\\caption{", caption, "}"), paste0("\\label{", label, "}"),
               paste0("\\begin{tabular}{", align, "}"), "\\toprule",
```

```
              paste0(paste(names(df), collapse = " & "), " \\\\"), "\\midrule",
              paste0(apply(df, 1, paste, collapse = " & "), " \\\\"), "\\bottomrule", "\\end{tabular}",
              "\\begin{tablenotes}\\footnotesize", paste0("\\item \\textit{Note}. ", note),
               ↪  "\\end{tablenotes}",
              "\\end{threeparttable}", "\\end{table}"), file.path(OUT, file))
}
LED <- list(); led <- function(k, v, note = "")
  LED[[length(LED) + 1]] <<- data.frame(key = k, value = as.character(v), note = note)
lv4 <- c("Only Prolife", "Prolife talking about Prochoice",
         "Prochoice talking about Prolife", "Only Prochoice")
code <- c("OPL", "PLPC", "PCPL", "OPC")
nice <- c(OPL = "Only Pro-life", PLPC = "Pro-life talking about Pro-choice",
          PCPL = "Pro-choice talking about Pro-life", OPC = "Only Pro-choice")
```

1 1. Data and sample

```
raw <- read.csv(p$data_path, stringsAsFactors = FALSE, colClasses = c(user = "character"))
f_cols <- c("care", "fairness", "loyalty", "authority", "purity")
stopifnot(all(c("Label", "sentiment_score", "user", "source", "prolife_prochoice", f_cols) %in% names(raw)))
for (v in f_cols) raw[[v]] <- as.integer(raw[[v]] %in% 1)
d <- raw[raw$Label %in% lv4 & !is.na(raw$sentiment_score), ]
d$cat <- factor(code[match(d$Label, lv4)], levels = code)
d$any_I <- as.integer(d$care == 1 | d$fairness == 1)
d$any_B <- as.integer(d$loyalty == 1 | d$authority == 1 | d$purity == 1)
d$fam_orig <- factor(ifelse(d$any_I == 1, "Individualizing",
                            ifelse(d$any_B == 1, "Binding", NA)), levels = c("Binding", "Individualizing"))
d$fam3 <- factor(ifelse(d$any_I == 1 & d$any_B == 1, "Both",
                        ifelse(d$any_I == 1, "Individualizing only", ifelse(d$any_B == 1, "Binding only",
                         ↪  NA))),
                 levels = c("Binding only", "Individualizing only", "Both"))
d$sent <- d$sentiment_score; SM <- mean(d$sent); d$sent_c <- d$sent - SM
d$.cl <- ifelse(is.na(d$user) | d$user == "", paste0("na_", seq_len(nrow(d))), d$user)
d$.row <- seq_len(nrow(d))
d$source_f <- factor(ifelse(is.na(d$source) | d$source == "", "(missing)", d$source))
d$agree <- as.integer((d$cat %in% c("OPL", "PLPC") & d$prolife_prochoice %in% "life")
                      | (d$cat %in% c("PCPL", "OPC") & d$prolife_prochoice %in% "choice"))

say("Rows in file: ", f_num(nrow(raw)), "; 'Neither': ", f_num(sum(raw$Label == "Neither", na.rm = TRUE)),
    "; analytic sample: ", f_num(nrow(d)), "; posts with no foundation tagged: ",
    f_num(sum(d$any_I == 0 & d$any_B == 0)),
    "; missing sentiment dropped: ", f_num(sum(raw$Label %in% lv4 & is.na(raw$sentiment_score))))
```

```
Rows in file: 199,833; 'Neither': 29,590; analytic sample: 170,243; posts with no foundation tagged: 35;
 ↪  missing sentiment dropped: 0
```

```
say("Clusters (user column as recorded): ", f_num(n_distinct(d$.cl)),
    "; user IDs in scientific notation: ",
    f_pct(mean(grepl("[eE]\\+", d$user))), "%; sentiment mean = ", f_d(SM, 3))
```

```
Clusters (user column as recorded): 111,931; user IDs in scientific notation: 29.5%; sentiment mean = -0.098
```

```
led("n_classified", f_num(nrow(raw))); led("n_neither", f_num(sum(raw$Label ==
                                                                  "Neither", na.rm = TRUE)));
                                                                   ↪  led("n_analytic", f_num(nrow(d)))
led("n_clusters", f_num(n_distinct(d$.cl)), "rounded IDs merge some accounts")

desc <- d %>% group_by(cat) %>% summarise(n = n(), any_I = mean(any_I), any_B = mean(any_B),
  I_only = mean(fam3 %in% "Individualizing only"), B_only = mean(fam3 %in% "Binding only")
  , Both = mean(fam3 %in% "Both"),
  care = mean(care), fairness = mean(fairness), loyalty = mean(loyalty),
  authority = mean(authority), purity = mean(purity),
  sent_mean = mean(sent), sent_sd = sd(sent), .groups = "drop")
```

```
block("Composition by category (proportions of posts)",
      desc %>% mutate(across(where(is.double), ~ round(.x, 3))))
```

```
### Composition by category (proportions of posts)
  cat     n any_I any_B I_only B_only  Both  care fairness loyalty authority purity sent_mean sent_sd
  OPL 41116 0.940 0.562  0.437  0.060 0.502 0.861    0.375   0.144     0.186  0.393     0.036   0.596
 PLPC 32940 0.970 0.534  0.465  0.030 0.505 0.903    0.433   0.129     0.099  0.425    -0.165   0.595
 PCPL 62719 0.978 0.574  0.426  0.022 0.552 0.771    0.777   0.334     0.106  0.241    -0.231   0.576
  OPC 33468 0.983 0.370  0.630  0.017 0.353 0.803    0.785   0.181     0.133  0.097     0.054   0.595
```

```
for (i in seq_len(nrow(desc))) { k <- as.character(desc$cat[i])
  for (v in c("any_I", "any_B", "B_only", "Both", "loyalty", "authority", "purity"))
    led(paste0("raw_pct_", v, "_", k), f_pct(desc[[v]][i]))
  led(paste0("sent_mean_", k), f_d(desc$sent_mean[i])); led(paste0("n_", k), f_num(desc$n[i])) }
block("Category x source (counts)",
      as.data.frame.matrix(table(d$cat, d$source_f)) %>% mutate(cat = code, .before = 1))
```

```
### Category x source (counts)
  cat   isi mortezas
  OPL 13177    27939
 PLPC 11202    21738
 PCPL 50794    11925
  OPC 26252     7216
```

```
write_tex(desc %>% transmute(Category = nice[as.character(cat)], `
                             $N$` = t_num(n), `Any I` = f_pct(any_I), `Any B` = f_pct(any_B),
                             `I only` = f_pct(I_only),
                             `B only` = f_pct(B_only), Both = f_pct(Both), Care = f_pct(care), Fairness =
                              ↪  f_pct(fairness),
                             Loyalty = f_pct(loyalty), Authority = f_pct(authority), Purity = f_pct(purity),
                             Sentiment = sprintf("%s (%s)", f_d(sent_mean), f_d(sent_sd))),
          "tab_s1_composition.tex",
          "Moral-foundation use by framing category (percentage of posts).", "tab:s1_composition",
          paste0("I = individualizing (care, fairness);
                 B = binding (loyalty, authority, purity). Families are not mutually exclusive.
                 Sentiment is the mean (SD) VADER compound score. $N = ", t_num(nrow(d)), "$."),
          size = "\\scriptsize")
```

2 2. Primary model

```
fit_stack <- function(dd, rhs = c("cat", "sent_c")) {
  long <- bind_rows(mutate(dd, .y = any_I, .out = "I"), mutate(dd, .y = any_B, .out = "B"))
  long$.out <- factor(long$.out, levels = c("I", "B"))
  fit <- glm(as.formula(paste(".y ~ 0 + .out + .out:(", paste(rhs, collapse = " + "), ")")), binomial, data =
 ↪  long)
  list(fit = fit, V = vcovCL(fit, cluster = long$.cl, type = "HC1"), n = nrow(dd))
}
cn <- function(o, k) paste0(".out", o, ":cat", k)
ws <- function(...) { a <- list(...); nm <- unique(unlist(lapply(a, names)));
o <- setNames(numeric(length(nm)), nm); for (w in a) o[names(w)] <- o[names(w)] + w; o }
W <- list(PL_I = setNames(1, cn("I", "PLPC")), PC_I = setNames(c(1, -1), cn("I", c("PCPL", "OPC"))),
          PL_B = setNames(1, cn("B", "PLPC")), PC_B = setNames(c(1, -1), cn("B", c("PCPL", "OPC"))))
W$ASY_B <- ws(W$PC_B, W$PL_B); W$ASY_OPP <- ws(W$PC_B, -W$PL_I)
lab <- c(PL_I = "Pro-life shift, individualizing", PC_I =
           "Pro-choice shift, individualizing", PL_B = "Pro-life shift, binding",
         PC_B = "Pro-choice shift, binding", ASY_B = "Convergence asymmetry, binding (+ = pro-choice more)",
         ASY_OPP = "Convergence toward opponent's family: PC binding minus PL individualizing (+ = pro-choice
          ↪  more)")
lc <- function(m, w) { b <- coef(m$fit); e <- sum(w * b[names(w)]);
s <- sqrt(drop(t(w) %*% m$V[names(w), names(w)] %*% w));
```

```
c(est = e, se = s, z = e / s, p = 2 * pnorm(-abs(e / s))) }
con <- function(m, spec) bind_rows(lapply(names(W),
                                          function(k) { r <- lc(m, W[[k]]);
                                          data.frame(spec = spec, contrast = k, est = r[[1]],
                                                     se = r[[2]], z = r[[3]], p = r[[4]]) }))

P <- fit_stack(d); pc <- con(P, "Primary")
block("Primary contrasts (log odds; CR1 SEs)", pc %>% mutate(label = lab[contrast],
                                                             OR = round(exp(est), 3), CI = f_or(est, se),
                                                           est = round(est, 3),
                                                           se = round(se, 4),
                                                           z = round(z, 2), p = f_p(p)))
```

```
### Primary contrasts (log odds; CR1 SEs)
    spec contrast    est     se     z      p
     ↪  label    OR               CI
 Primary     PL_I  0.615 0.0410 14.98 < .001
  ↪  Pro-life shift, individualizing 1.849 1.85 [1.71, 2.00]
 Primary     PC_I -0.463 0.0592 -7.82 < .001
  ↪  Pro-choice shift, individualizing 0.629 0.63 [0.56, 0.71]
 Primary     PL_B -0.064 0.0233 -2.75 = .006
  ↪  Pro-life shift, binding 0.938 0.94 [0.90, 0.98]
 Primary     PC_B  0.903 0.0152 59.39 < .001
  ↪  Pro-choice shift, binding 2.467 2.47 [2.39, 2.54]
 Primary    ASY_B  0.839 0.0265 31.63 < .001                                                    Convergence asymmetry,
  ↪  binding (+ = pro-choice more) 2.314 2.31 [2.20, 2.44]
 Primary  ASY_OPP  0.288 0.0443  6.51 < .001 Convergence toward opponent's family: PC binding minus PL
  ↪  individualizing (+ = pro-choice more) 1.334 1.33 [1.22, 1.46]
```

```
for (i in seq_len(nrow(pc))) led(paste0("test_", pc$contrast[i]),
                                 sprintf("OR = %s; b = %s, SE = %s, z = %s, p %s",
                                         f_or(pc$est[i], pc$se[i]),
                                 f_d(pc$est[i]), f_d(pc$se[i]), f_d(pc$z[i]),
                                 f_p(pc$p[i])), lab[pc$contrast[i]])

# Predicted shares at mean sentiment and percentage-point shifts (delta method)
pw <- function(o, k) { w <- setNames(1, paste0(".out", o));
if (k != "OPL") w <- ws(w, setNames(1, cn(o, k))); w }
b <- coef(P$fit)
pred <- bind_rows(lapply(c("I", "B"), function(o) bind_rows(lapply(code, function(k)
  { w <- pw(o, k); e <- sum(w * b[names(w)])
  s <- sqrt(drop(t(w) %*% P$V[names(w), names(w)] %*% w));
  data.frame(out = o, cat = k, p = plogis(e), lo = plogis(e - 1.96 * s), hi = plogis(e + 1.96 * s)) }))))
block("Predicted % of posts (sentiment at mean)",
      pred %>% mutate(across(c(p, lo, hi), ~ round(100 * .x, 1))))
```

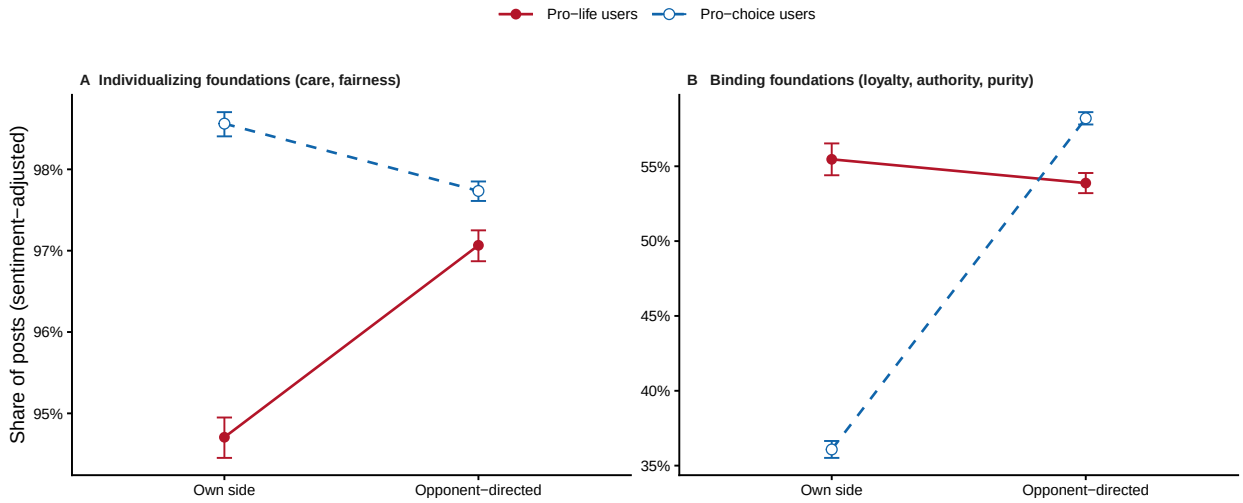
```
### Predicted % of posts (sentiment at mean)
 out  cat    p   lo   hi
   I  OPL 94.7 94.5 94.9
   I PLPC 97.1 96.9 97.2
   I PCPL 97.7 97.6 97.9
   I  OPC 98.6 98.4 98.7
   B  OPL 55.5 54.4 56.5
   B PLPC 53.9 53.2 54.5
   B PCPL 58.2 57.8 58.6
   B  OPC 36.1 35.5 36.6
```

```
for (i in seq_len(nrow(pred))) led(paste0("adj_pct_", pred$out[i], "_", pred$cat[i]),
                                   f_pct(pred$p[i]), sprintf("[%s, %s]",
                                                             f_pct(pred$lo[i]), f_pct(pred$hi[i])))
pdiff <- function(o, k1, k2) { w1 <- pw(o, k1); w2 <- pw(o, k2);
p1 <- plogis(sum(w1 * b[names(w1)])); p2 <- plogis(sum(w2 * b[names(w2)]))
  g <- ws(p2 * (1 - p2) * w2, -p1 * (1 - p1) * w1);
  s <- sqrt(drop(t(g) %*% P$V[names(g), names(g)] %*% g)); c(d = p2 - p1, s = s) }
for (o in c("I", "B")) for (sd in list(c("Pro-life", "OPL", "PLPC"),
```

```
                                    c("Pro-choice", "OPC", "PCPL"))) {
  r <- pdiff(o, sd[2], sd[3]); led(paste0("pp_shift_", o, "_", sd[1]),
                                   sprintf("%s [%s, %s]", f_d(100 * r[["d"]], 1),
      f_d(100 * (r[["d"]] - 1.96 * r[["s"]]), 1),
      f_d(100 * (r[["d"]] + 1.96 * r[["s"]]), 1)), "percentage points") }
gp <- function(o, k) pred$p[pred$out == o & pred$cat == k]
led("headroom_closed_PL_I", f_pct((gp("I", "PLPC") - gp("I", "OPL")) / (1 - gp("I", "OPL"))),
    "share of distance to 100% closed (descriptive)")
led("headroom_closed_PC_B", f_pct((gp("B", "PCPL") - gp("B", "OPC")) / (1 - gp("B", "OPC"))),
    "share of distance to 100% closed (descriptive)")

tp <- pc %>% transmute(Contrast = c("Pro-life: opponent-directed vs.\\ own side,
                                    individualizing", "Pro-choice: opponent-directed vs.\\ own side,
 ↪  individualizing",
                                    "Pro-life: opponent-directed vs.\\ own side, binding",
                                    "Pro-choice: opponent-directed vs.\\ own side, binding",
                                    "Convergence asymmetry, binding", "Convergence toward opponent's family"),
                       `$b$` = f_d(est, 3), SE = f_d(se, 3), `OR [95\\% CI]` = f_or(est, se), `$z$` = f_d(z),
                        ↪  `$p$` = t_p(p))
write_tex(tp, "tab_s1_primary.tex", "Shifts in moral-foundation use from self-focused to opponent-directed
 ↪  posts.", "tab:s1_primary",
          paste0("Logistic regressions predicting whether a post invokes any individualizing or any binding
           ↪  foundation from framing category and mean-centered sentiment, estimated jointly. ",
                 "Convergence asymmetry: pro-choice minus pro-life convergence toward the opponent's
                  ↪  characteristic family (positive = pro-choice users converge more); the binding row
                  ↪  compares both sides on binding use, the last row compares pro-choice binding with
                  ↪  pro-life individualizing. For these rows the OR column is a ratio of odds ratios. ",
                 "CR1 SEs clustered by user. $N = ", t_num(P$n), "$ posts."), align = "lrrrrr")
```

```
fd <- pred %>% mutate(panel = factor(ifelse(out == "I",
                                            "A  Individualizing foundations (care, fairness)",
                                            "B   Binding foundations (loyalty, authority, purity)")),
  target = factor(ifelse(cat %in% c("OPL", "OPC"), "Own side", "Opponent-directed"),
                  levels = c("Own side", "Opponent-directed")),
  side = factor(ifelse(cat %in% c("OPL", "PLPC"), "Pro-life users", "Pro-choice users"),
                levels = c("Pro-life users", "Pro-choice users")))
g <- ggplot(fd, aes(target, 100 * p, group = side, colour = side, shape = side, linetype = side)) +
  geom_line(linewidth = 0.5) + geom_errorbar(aes(ymin = 100 * lo, ymax = 100 * hi), width = 0.06, linetype =
   ↪  "solid") +
  geom_point(size = 1.8, fill = "white") + facet_wrap(~ panel, scales = "free_y") +
  scale_colour_manual(values = c("Pro-life users" = "#B31529", "Pro-choice users" = "#1065AB")) +
  scale_shape_manual(values = c("Pro-life users" = 16, "Pro-choice users" = 21)) +
  scale_linetype_manual(values = c("Pro-life users" = "solid", "Pro-choice users" = "dashed")) +
  scale_y_continuous(labels = function(x) paste0(x, "%")) +
  labs(x = NULL, y = "Share of posts (sentiment-adjusted)", colour = NULL, shape = NULL, linetype = NULL) +
  theme_classic(base_size = 8) + theme(legend.position = "top", strip.background = element_blank()
                                       , strip.text = element_text(hjust = 0, face = "bold"))
print(g); ggsave(file.path(OUT, "fig1_study1.pdf"), g, width = 7, height = 2.9)
```




```
write.csv(pred, file.path(OUT, "fig1_data.csv"), row.names = FALSE)
```

3 3. Individual foundations

```
ft <- bind_rows(lapply(f_cols, function(v) { dd <- d; dd$.y <- dd[[v]]
  g <- glm(.y ~ cat + sent_c, binomial, dd); V <- vcovCL(g, cluster = dd$.cl, type = "HC1"); bb <- coef(g)
  sh <- function(w) { e <- sum(w * bb[names(w)]); s <- sqrt(drop(t(w) %*% V[names(w), names(w)] %*% w));
  c(e, s, 2 * pnorm(-abs(e / s))) }
  pl <- sh(setNames(1, "catPLPC")); pcs <- sh(setNames(c(1, -1), c("catPCPL", "catOPC")))
  pr <- plogis(bb[["(Intercept)"]] + c(0, bb[["catPLPC"]], bb[["catPCPL"]], bb[["catOPC"]]))
  data.frame(foundation = v, OPL = 100 * pr[1], PLPC = 100 * pr[2], PCPL = 100 * pr[3], OPC = 100 * pr[4],
             PL_OR = f_or(pl[1], pl[2]), PL_p = f_p(pl[3]), PC_OR = f_or(pcs[1], pcs[2]), PC_p = f_p(pcs[3]))
              ↪  }))
block("Per-foundation predicted % and shift ORs (CR1)", ft %>% mutate(across(OPL:OPC, ~ round(.x, 1))))
```

```
### Per-foundation predicted % and shift ORs (CR1)
 foundation  OPL PLPC PCPL  OPC            PL_OR   PL_p            PC_OR   PC_p
       care 87.5 90.4 76.3 82.2 1.34 [1.26, 1.43] < .001 0.70 [0.67, 0.72] < .001
   fairness 36.8 43.6 78.2 78.1 1.32 [1.26, 1.39] < .001 1.01 [0.97, 1.05] = .605
    loyalty 13.6 13.0 34.4 16.9 0.95 [0.90, 1.00] = .066 2.57 [2.48, 2.67] < .001
  authority 14.5  8.9 10.4  9.9 0.58 [0.54, 0.61] < .001 1.05 [1.01, 1.11] = .027
     purity 39.8 42.2 23.6  9.9 1.10 [1.04, 1.17] = .001 2.83 [2.71, 2.95] < .001
```

```
for (i in seq_len(nrow(ft))) { led(paste0("OR_PL_shift_",
                                          ft$foundation[i]), ft$PL_OR[i], paste("p", ft$PL_p[i]));
  led(paste0("OR_PC_shift_", ft$foundation[i]), ft$PC_OR[i], paste("p", ft$PC_p[i])) }
write_tex(ft %>% transmute(Foundation = tools::toTitleCase(foundation),
                           `Only PL` = f_d(OPL, 1), `PL$\\rightarrow$PC` = f_d(PLPC, 1),
                           `PC$\\rightarrow$PL` = f_d(PCPL, 1), `Only PC` = f_d(OPC, 1),
                           `Pro-life shift OR [95\\% CI]` = PL_OR, `Pro-choice shift OR [95\\% CI]` = PC_OR),
          "tab_s1_foundations.tex", "Use of each moral foundation by framing category.", "tab:s1_foundations",
          "Columns 2--5: sentiment-adjusted percentage of posts invoking the foundation.
          Shift ORs compare opponent-directed with self-focused posts within each side; CR1 SEs clustered by
 ↪  user.",
          size = "\\scriptsize")
```

4 4. Original multinomial model and alternative coding

```
fit_mn <- function(f, data) { dd <- data[complete.cases(data[, all.vars(f)]), ];
list(fit = multinom(f, dd, Hess = TRUE, maxit = 1000, trace = FALSE), data = dd) }
mn_b <- function(m) { C <- coef(m); setNames(as.vector(t(C)), as.vector(t(outer(rownames(C),
                                                                                colnames(C), paste, sep =
                                                                                 ↪  ":")))) }
mn_V <- function(o, type = "cluster") { m <- o$fit; dd <- o$data; B <- vcov(m); if (type == "model") return(B)
  X <- model.matrix(delete.response(terms(m)), dd); Y <- model.matrix(~ cat - 1, dd); R <- (Y - fitted(m))[,
 ↪  -1, drop = FALSE]
  S <- do.call(cbind, lapply(seq_len(ncol(R)), function(k) R[, k] * X));
  colnames(S) <- as.vector(t(outer(m$lab[-1], colnames(X), paste, sep = ":")))
  S <- S[, colnames(B)]; Sc <- rowsum(S, dd$.cl); G <- nrow(Sc); n <- nrow(S); k <- ncol(S)
  G / (G - 1) * (n - 1) / (n - k) * B %*% crossprod(Sc) %*% B }
# Convergence asymmetry for an I-vs-B predictor (+ = pro-choice converges more): -(b[PLPC] + b[PCPL] - b[OPC])
asy <- function(bb, V, term, s = NULL) { w <- setNames(c(-1, -1, 1), paste0(c("PLPC:", "PCPL:", "OPC:"),
 ↪  term))
  if (!is.null(s)) w <- ws(w, setNames(s * c(-1, -1, 1), paste0(c("PLPC:", "PCPL:", "OPC:"), term, ":sent")))
  e <- sum(w * bb[names(w)]); se <- sqrt(drop(t(w) %*% V[names(w),
                                                         names(w)] %*% w));
  c(est = e, se = se, z = e / se, p = 2 * pnorm(-abs(e / se))) }

M1 <- fit_mn(cat ~ fam_orig + sent, d); M2 <- fit_mn(cat ~ fam_orig * sent, d);
M3 <- fit_mn(cat ~ fam3 + sent, d)
b1 <- mn_b(M1$fit); b2 <- mn_b(M2$fit); b3 <- mn_b(M3$fit);
V1 <- mn_V(M1); V2 <- mn_V(M2); V3 <- mn_V(M3)
lr <- as.data.frame(anova(M1$fit, M2$fit)); nc <- ncol(lr)
say("\nOriginal multinomial N = ", f_num(nrow(M1$data)), "; Individualizing coefficients (M1): ",
    paste(round(b1[paste0(c("PLPC", "PCPL", "OPC"), ":fam_origIndividualizing")], 4), collapse = ", "),
    " (original output: 0.6155, 0.8851, 1.3418);
    interaction LR chi2(", lr[2, nc - 2], ") = ", round(lr[2, nc - 1], 2))
```

```
Original multinomial N = 170,208; Individualizing coefficients (M1): 0.6155, 0.8851, 1.3418 (original output:
 ↪  0.6155, 0.8851, 1.3418);
    interaction LR chi2(3) = 149.97
```

```
m2 <- data.frame(term = names(b2), b = b2, se_model = sqrt(diag(mn_V(M2, "model")))[names(b2)],
                 se_CR1 = sqrt(diag(V2))[names(b2)])
block("Original interaction model (M2): b, model-based SE, CR1 SE",
      m2 %>% mutate(across(where(is.numeric), ~ round(.x, 4))))
```

```
### Original interaction model (M2): b, model-based SE, CR1 SE
                             term       b se_model se_CR1
                 PLPC:(Intercept) -0.7442   0.0399 0.0416
      PLPC:fam_origIndividualizing  0.5138   0.0407 0.0428
                        PLPC:sent -1.1021   0.0700 0.0717
 PLPC:fam_origIndividualizing:sent  0.5782   0.0711 0.0731
                 PCPL:(Intercept) -0.4163   0.0363 0.0392
      PCPL:fam_origIndividualizing  0.7994   0.0369 0.0392
                        PCPL:sent -1.3405   0.0641 0.0689
 PCPL:fam_origIndividualizing:sent  0.6266   0.0651 0.0697
                  OPC:(Intercept) -1.3340   0.0496 0.0607
       OPC:fam_origIndividualizing  1.1701   0.0502 0.0608
                         OPC:sent -0.6643   0.0858 0.0837
  OPC:fam_origIndividualizing:sent  0.7626   0.0867 0.0847
```

```
m2t <- m2 %>% mutate(cat = sub(":.*", "", term), pred = sub("^[^:]+:", "", term),
                     cell = sprintf("%s (%s)", f_d(b, 3), f_d(se_CR1, 3))) %>%
  mutate(pred = recode(pred, `(Intercept)` = "Intercept",
                       fam_origIndividualizing = "Individualizing (vs.\\ binding only)",
                       sent = "Sentiment", `fam_origIndividualizing:sent` = "Individualizing $\\times$
                        ↪  sentiment")) %>%
  select(pred, cat, cell) %>% pivot_wider(names_from = cat, values_from = cell)
```

```
names(m2t) <- c("Predictor", "PL$\\rightarrow$PC", "PC$\\rightarrow$PL", "Only PC")
write_tex(m2t, "tab_s1_multinomial.tex", "Original specification: multinomial logistic regression predicting
↪  framing category.", "tab:s1_multinomial",
          paste0("Reference: Only Pro-life. Cells are $b$ (CR1 SE, clustered by user).
                  A post is coded individualizing if it invokes care or fairness and binding only if it invokes
↪  loyalty, authority, or purity and no individualizing foundation; posts invoking both families are
↪  therefore coded individualizing. Sentiment is the raw VADER compound score. $N = ",
                  t_num(nrow(M2$data)), "$."), align = "lrrr")

alt <- bind_rows(
  data.frame(spec = "Original coding (both-family posts = individualizing), main effects", t(asy(b1, V1,
   ↪  "fam_origIndividualizing"))),
  data.frame(spec = "Original coding, interaction model, at mean sentiment", t(asy(b2, V2,
   ↪  "fam_origIndividualizing", SM))),
  data.frame(spec = "Recoded: individualizing only vs. binding only", t(asy(b3, V3, "fam3Individualizing
   ↪  only"))),
  data.frame(spec = "Recoded: both families vs. binding only", t(asy(b3, V3, "fam3Both"))),
  pc %>% filter(contrast == "ASY_B") %>% transmute(spec = "Primary: any binding foundation", est, se, z, p),
  pc %>% filter(contrast == "ASY_OPP") %>% transmute(spec = "Primary: toward opponent's family", est, se, z,
↪  p))
block("Convergence asymmetry across codings (+ = pro-choice converges more; CR1)", alt %>% mutate(est =
↪  round(est, 3), se = round(se, 4), z = round(z, 2), p = f_p(p)))
```

```
### Convergence asymmetry across codings (+ = pro-choice converges more; CR1)
                                                                spec    est     se     z      p
 Original coding (both-family posts = individualizing), main effects -0.159 0.0724 -2.19 = .028
              Original coding, interaction model, at mean sentiment -0.100 0.0765 -1.31 = .191
                    Recoded: individualizing only vs. binding only  0.264 0.0732  3.61 < .001
                           Recoded: both families vs. binding only -0.626 0.0745 -8.40 < .001
                                   Primary: any binding foundation  0.839 0.0265 31.63 < .001
                                  Primary: toward opponent's family  0.288 0.0443  6.51 < .001
```

```
for (i in seq_len(nrow(alt))) led(paste0("asy_", i), sprintf("b = %s, SE = %s, z = %s, p %s", f_d(alt$est[i],
↪  3), f_d(alt$se[i], 3), f_d(alt$z[i]), f_p(alt$p[i])), alt$spec[i])
write_tex(alt %>% transmute(Specification = spec, `$b$` = f_d(est, 3), SE = f_d(se, 3), `$z$` = f_d(z), `$p$`
↪  = t_p(p)),
          "tab_s1_altspec.tex", "Convergence asymmetry under alternative codings of moral-foundation use.",
           ↪  "tab:s1_altspec",
          "Positive values: pro-choice users converge more toward the opponent's characteristic family than
           ↪  pro-life users; negative values: the reverse. CR1 SEs clustered by user.",
          align = "lrrrr", size = "\\scriptsize")
```

5 5. Two robustness checks

```
rob <- bind_rows(pc %>% mutate(spec = "Primary"),
                 con(fit_stack(d, c("cat", "sent_c", "source_f")), "+ data source covariate"),
                 { dd <- d[d$agree == 1, ]; dd$sent_c <- dd$sent - mean(dd$sent);
                 con(fit_stack(dd), paste0("BERTweet and GPT stance agree (n = ", f_num(nrow(dd)), ")")) })
rw <- rob %>% mutate(cell = ifelse(grepl("^ASY", contrast), sprintf("%s (p %s)", f_d(est), f_p(p)), f_or(est,
↪  se))) %>%
  select(spec, contrast, cell) %>% pivot_wider(names_from = contrast, values_from = cell)
block("Robustness: shift ORs [95% CI]; asymmetries as log-odds (p)", rw)
```

```
### Robustness: shift ORs [95% CI]; asymmetries as log-odds (p)
                                      spec              PL_I              PC_I              PL_B
                                       ↪  PC_B           ASY_B         ASY_OPP
                                    Primary 1.85 [1.71, 2.00] 0.63 [0.56, 0.71] 0.94 [0.90, 0.98] 2.47 [2.39,
                                     ↪  2.54] 0.84 (p < .001) 0.29 (p < .001)
                        + data source covariate 1.84 [1.70, 1.99] 0.63 [0.56, 0.71] 0.94 [0.90, 0.98] 2.47 [2.39,
                         ↪  2.54] 0.84 (p < .001) 0.29 (p < .001)
 BERTweet and GPT stance agree (n = 152,312) 1.95 [1.79, 2.12] 0.73 [0.65, 0.83] 0.94 [0.89, 0.98] 2.47 [2.39,
  ↪  2.55] 0.84 (p < .001) 0.24 (p < .001)
```

```
write_tex(rw %>% rename(Specification = spec, `PL ind.` = PL_I, `PC ind.` = PC_I,
                        `PL bind.` = PL_B, `PC bind.` = PC_B, `Asym.\\ bind.` = ASY_B, `Asym.\\ opp.` =
                         ↪  ASY_OPP) %>%
            mutate(across(-1, ~ gsub("p < .001", "$p<.001$", gsub("p = ", "$p=$", .x)))),
          "tab_s1_robustness.tex", "Robustness of Study~1 shifts and asymmetries.", "tab:s1_robustness",
          "Shift columns: ORs [95\\% CI], opponent-directed vs.\\ self-focused posts within each side (ind. =
           ↪  any individualizing; bind. = any binding). Asymmetry columns: log-odds difference in convergence
           ↪  (positive = pro-choice users converge more). CR1 SEs clustered by user.",
          size = "\\scriptsize")
```

6 6. Numbers ledger

```
L <- bind_rows(LED); write.csv(L, file.path(OUT, "study1_numbers.csv"), row.names = FALSE)
block("NUMBERS LEDGER", L)
```

```
### NUMBERS LEDGER
                    key                                                                value
                     ↪  note
           n_classified                                                          199,833
              n_neither                                                           29,590
             n_analytic                                                          170,243
             n_clusters                                                          111,931
              ↪  rounded IDs merge some accounts
      raw_pct_any_I_OPL                                                            94.0
      raw_pct_any_B_OPL                                                            56.2
     raw_pct_B_only_OPL                                                             6.0
       raw_pct_Both_OPL                                                            50.2
    raw_pct_loyalty_OPL                                                            14.4
  raw_pct_authority_OPL                                                            18.6
     raw_pct_purity_OPL                                                            39.3
          sent_mean_OPL                                                            0.04
                  n_OPL                                                          41,116
     raw_pct_any_I_PLPC                                                            97.0
     raw_pct_any_B_PLPC                                                            53.4
    raw_pct_B_only_PLPC                                                             3.0
      raw_pct_Both_PLPC                                                            50.5
   raw_pct_loyalty_PLPC                                                            12.9
 raw_pct_authority_PLPC                                                             9.9
    raw_pct_purity_PLPC                                                            42.5
         sent_mean_PLPC                                                           -0.17
                 n_PLPC                                                          32,940
     raw_pct_any_I_PCPL                                                            97.8
     raw_pct_any_B_PCPL                                                            57.4
    raw_pct_B_only_PCPL                                                             2.2
      raw_pct_Both_PCPL                                                            55.2
   raw_pct_loyalty_PCPL                                                            33.4
 raw_pct_authority_PCPL                                                            10.6
    raw_pct_purity_PCPL                                                            24.1
         sent_mean_PCPL                                                           -0.23
                 n_PCPL                                                          62,719
      raw_pct_any_I_OPC                                                            98.3
      raw_pct_any_B_OPC                                                            37.0
     raw_pct_B_only_OPC                                                             1.7
       raw_pct_Both_OPC                                                            35.3
    raw_pct_loyalty_OPC                                                            18.1
  raw_pct_authority_OPC                                                            13.3
     raw_pct_purity_OPC                                                             9.7
          sent_mean_OPC                                                            0.05
                  n_OPC                                                          33,468
              test_PL_I  OR = 1.85 [1.71, 2.00]; b = 0.61, SE = 0.04, z = 14.98, p < .001
               ↪  Pro-life shift, individualizing
              test_PC_I OR = 0.63 [0.56, 0.71]; b = -0.46, SE = 0.06, z = -7.82, p < .001
               ↪  Pro-choice shift, individualizing
```

```
           test_PL_B OR = 0.94 [0.90, 0.98]; b = -0.06, SE = 0.02, z = -2.75, p = .006
            ↪  Pro-life shift, binding
           test_PC_B  OR = 2.47 [2.39, 2.54]; b = 0.90, SE = 0.02, z = 59.39, p < .001
            ↪  Pro-choice shift, binding
          test_ASY_B  OR = 2.31 [2.20, 2.44]; b = 0.84, SE = 0.03, z = 31.63, p < .001
           ↪  Convergence asymmetry, binding (+ = pro-choice more)
        test_ASY_OPP   OR = 1.33 [1.22, 1.46]; b = 0.29, SE = 0.04, z = 6.51, p < .001 Convergence toward
         ↪  opponent's family: PC binding minus PL individualizing (+ = pro-choice more)
       adj_pct_I_OPL                                                              94.7
        ↪  [94.5, 94.9]
      adj_pct_I_PLPC                                                              97.1
       ↪  [96.9, 97.2]
      adj_pct_I_PCPL                                                              97.7
       ↪  [97.6, 97.9]
       adj_pct_I_OPC                                                              98.6
        ↪  [98.4, 98.7]
       adj_pct_B_OPL                                                              55.5
        ↪  [54.4, 56.5]
      adj_pct_B_PLPC                                                              53.9
       ↪  [53.2, 54.5]
      adj_pct_B_PCPL                                                              58.2
       ↪  [57.8, 58.6]
       adj_pct_B_OPC                                                              36.1
        ↪  [35.5, 36.6]
  pp_shift_I_Pro-life                                          2.4 [2.1, 2.7]
   ↪  percentage points
pp_shift_I_Pro-choice                                        -0.8 [-1.0, -0.6]
 ↪  percentage points
  pp_shift_B_Pro-life                                        -1.6 [-2.7, -0.5]
   ↪  percentage points
pp_shift_B_Pro-choice                                        22.1 [21.4, 22.8]
 ↪  percentage points
 headroom_closed_PL_I                                                         44.6
  ↪  share of distance to 100% closed (descriptive)
 headroom_closed_PC_B                                                         34.6
  ↪  share of distance to 100% closed (descriptive)
     OR_PL_shift_care                                        1.34 [1.26, 1.43]
      ↪  p < .001
     OR_PC_shift_care                                        0.70 [0.67, 0.72]
      ↪  p < .001
 OR_PL_shift_fairness                                        1.32 [1.26, 1.39]
  ↪  p < .001
 OR_PC_shift_fairness                                        1.01 [0.97, 1.05]
  ↪  p = .605
  OR_PL_shift_loyalty                                        0.95 [0.90, 1.00]
   ↪  p = .066
  OR_PC_shift_loyalty                                        2.57 [2.48, 2.67]
   ↪  p < .001
OR_PL_shift_authority                                        0.58 [0.54, 0.61]
 ↪  p < .001
OR_PC_shift_authority                                        1.05 [1.01, 1.11]
 ↪  p = .027
   OR_PL_shift_purity                                        1.10 [1.04, 1.17]
    ↪  p = .001
   OR_PC_shift_purity                                        2.83 [2.71, 2.95]
    ↪  p < .001
              asy_1                      b = -0.159, SE = 0.072, z = -2.19, p = .028
               ↪  Original coding (both-family posts = individualizing), main effects
              asy_2                      b = -0.100, SE = 0.076, z = -1.31, p = .191
               ↪  Original coding, interaction model, at mean sentiment
              asy_3                       b = 0.264, SE = 0.073, z = 3.61, p < .001
               ↪  Recoded: individualizing only vs. binding only
              asy_4                      b = -0.626, SE = 0.074, z = -8.40, p < .001
               ↪  Recoded: both families vs. binding only
              asy_5                       b = 0.839, SE = 0.027, z = 31.63, p < .001
               ↪  Primary: any binding foundation
              asy_6                       b = 0.288, SE = 0.044, z = 6.51, p < .001
               ↪  Primary: toward opponent's family
```

```r
say("\nFiles: ", paste(list.files(OUT), collapse = ", "))
```

```
Files: fig1_data.csv, fig1_study1.pdf, study1_numbers.csv, study1_results.txt, tab_s1_altspec.tex,
 ↪  tab_s1_composition.tex, tab_s1_foundations.tex, tab_s1_multinomial.tex, tab_s1_primary.tex,
 ↪  tab_s1_robustness.tex
```